

Heart Disease Risk Assessment System

SHAP Explainability Report

The goal of this analysis is to **interpret the machine learning model** used for predicting heart disease using SHAP (SHapley Additive exPlanations).

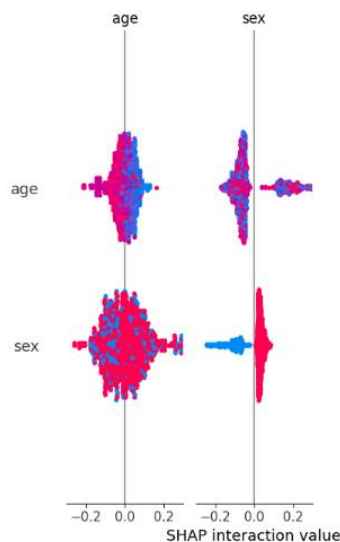
SHAP helps us understand **which features contribute most** to the prediction, both globally (all patients) and locally (individual patients).

Global Feature Importance

Global SHAP plots show **which features influence the model most across all patients**.

Bar Plot

- Shows **average impact of each feature** on the model output.
- Features with **higher bars** are more important.



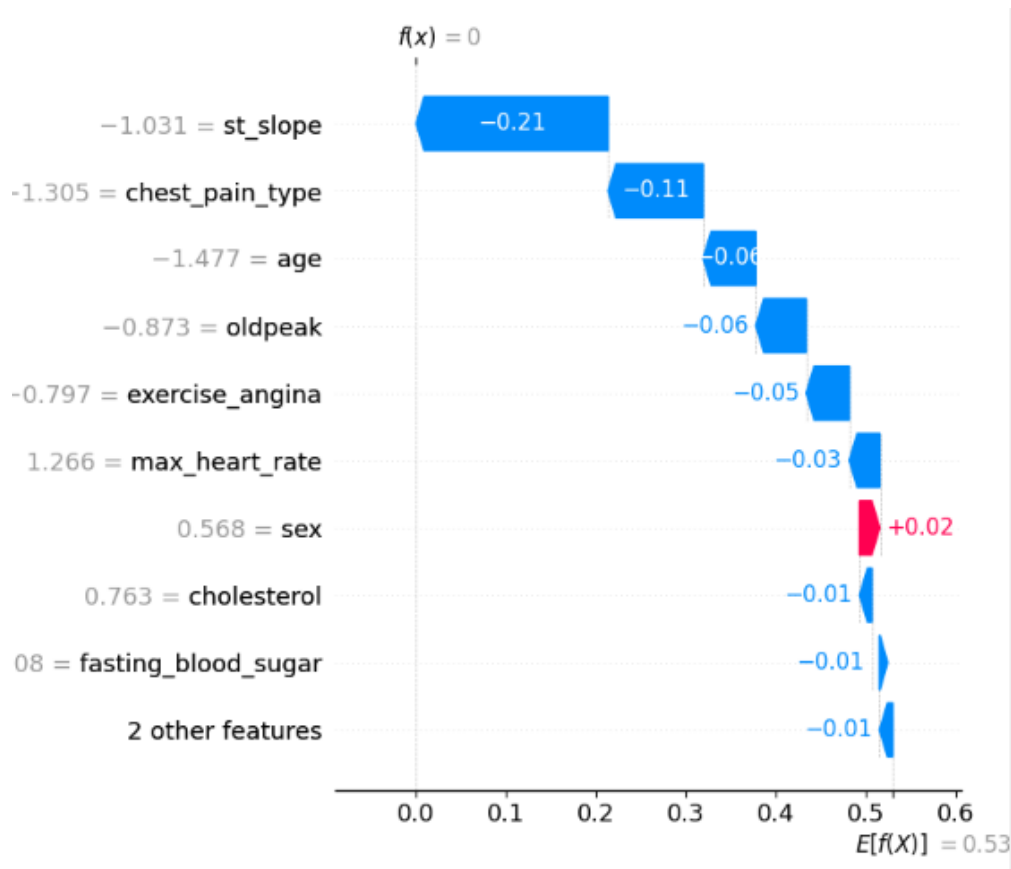
Dot Plot

- Shows **distribution of SHAP values** for each feature.
- Helps identify **positive or negative contributions** for individual samples.

Local Explanation

Local SHAP plots show **feature contributions for a single patient**.

- **Waterfall plot:** Visualizes how each feature increases or decreases the risk prediction.
- **Base value:** Average model output.
- **Prediction:** Final risk score for this patient.



Conclusion

- SHAP allows us to **understand both global and individual predictions** of the heart disease model.
- The analysis highlights which features drive risk predictions, **helping in clinical decision-making**.
- Using these visualizations, doctors or stakeholders can **trust and validate the model**.